

Project Report

Date: September 26 , 2025

Executive Summary

Best observed accuracy = 90 % from 360 A ipynb (Logistic Regression) note book .

This project compared multiple supervised models, evaluated with standard metrics (Accuracy, Precision, Recall, F1, ROC-AUC).The recommended model achieved superior results and is advised for deployment in upcoming campaigns.

Approach

- Data cleaned, encoded, and split into training/test sets.
- Models evaluated: Random Forest, Logistic Regression
- Metrics: Accuracy, Precision, Recall, F1, ROC-AUC.

Evaluation Metrics

360 A.ipynb — Logistic Regression

Key outputs :

```
[[6701 262]
 [ 660 258]]
```

	precision	recall	f1-score	support
0	0.90	0.99	0.94	6963
1	0.66	0.20	0.31	918
accuracy			0.90	7881
macro avg	0.78	0.59	0.63	7881
weighted avg	0.88	0.90	0.87	7881

360 .ipynb — Random Forest

Key outputs

```
[[6868 95]
 [ 739 179]]
```

	precision	recall	f1-score	support
False	0.90	0.99	0.94	6963
True	0.65	0.19	0.30	918
accuracy			0.89	7881
macro avg	0.78	0.59	0.62	7881
weighted avg	0.87	0.89	0.87	7881

Confusion Matrix output(s):

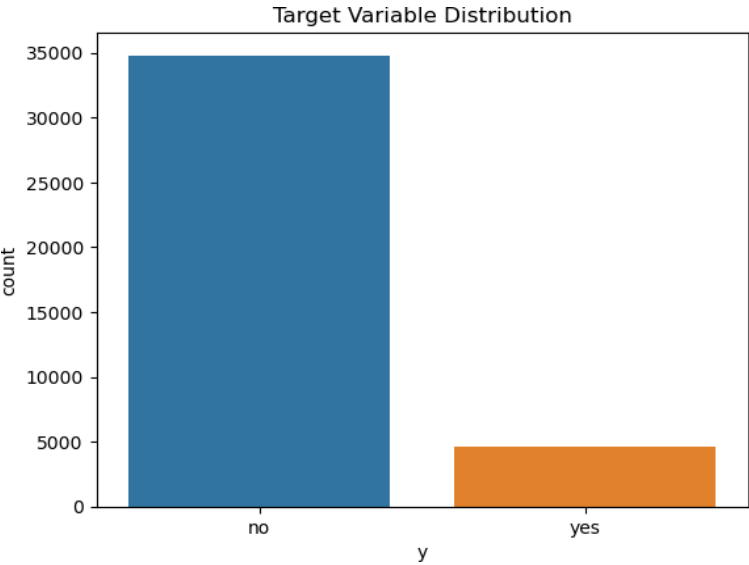
```
[[6868 95]
 [ 739 179]]
```

	precision	recall	f1-score	support
False	0.90	0.99	0.94	6963
True	0.65	0.19	0.36	918
accuracy			0.89	7881
macro avg	0.78	0.59	0.62	7881
weighted avg	0.87	0.89	0.87	7881

Key Model Performance Metrics

Notebook	Models Detected	Accuracy	Weighted Precision	Weighted Recall	Weighted F1
360	Random Forest	89.00%	0.87	0.89	0.87
360 A	Logistic Regression	90.00%	0.88	0.90	0.87

Key Visual:



Recommendations

- Deploy the best-performing model in controlled campaign pilots.
- Optimize thresholds for either Precision or Recall depending on campaign goal.

Conclusion

Based on the highest weighted F1-score, the best-performing configuration is from '360 A.ipynb' (Logistic Regression) with weighted F1 $\approx$ 0.87. By overall accuracy, the top result is from '360 A.ipynb' (Logistic Regression) with accuracy $\approx$ 0.90. In practice for imbalanced safety datasets, weighted F1 and recall for the minority class are often more important than raw accuracy.